PARTS LIST	
Semiconductors:	
Board1	- Arduino Uno board
Resistors (all 1/4-watt, ±5% carbon):	
R1-R3	- 1-kilo-ohm
VR1	- 10-kilo-ohm preset
Miscellaneous:	
LCD1	- 16x2 LCD
CON1	- 3-pin female connector for MOSFET
	- Jumper wires
	- MOSFET for testing

of Arduino UNO measure output voltage of the device. Note that, MOSFET (with front view) should be connected to the points marked 1, 2 and 3 in Fig. 2.

The authors' prototype of the breakout board where MOSFET can be connected is shown in Fig. 3. Based on the voltage received through analogue pins, Arduino identifies the type of MOSFET and the terminals corresponding to each pin of MOSFET.

For display purposes, Arduino sends the information to the LCD1 via digital pins 5, 4, 3 and 2 to LCD1 pins D4, D5, D6 and D7, respectively.

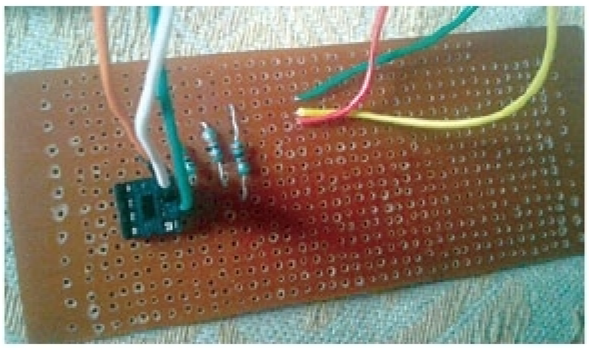

Fig. 3: Authors' prototype of breakout board where MOSFET can be connected

Note that, this project assumes source and body terminals of MOSFET to be internally shorted.

Software

The source code is written in Arduino programming language. ATmega328/328P is programmed using Arduino IDE software. Select the correct board from Tools→Board menu in Arduino IDE, and burn the program (sketch) through the standard USB port in your computer.

mosfet_find.ino code uses LiquidCrystal.h header file provided by Arduino library for working with LCD1.

EFY Note
The source code of this project is available for free download at *source.efymag.com*

lcd.begin(16, 2) function helps configure LCD1.

Serial.begin(9600) function initialises the serial port with a baud rate of 9600.

Construction and testing

Assemble all components as per the circuit diagram. Adjust the preset for full contrast of LCD1. Place MOSFET in connector CON1 for testing. EFY

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